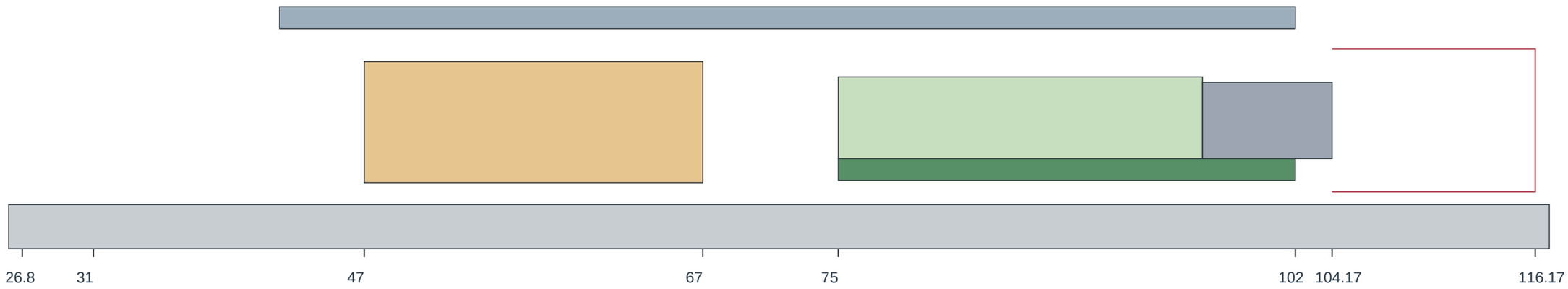
AXIAL-R1 SOURCE / REPAIRED ROTARY COLLAR INTEGRATION

Dimensions mm. Source board bytes preserved. Mechanical prototype; electrical and physical qualification remain open.

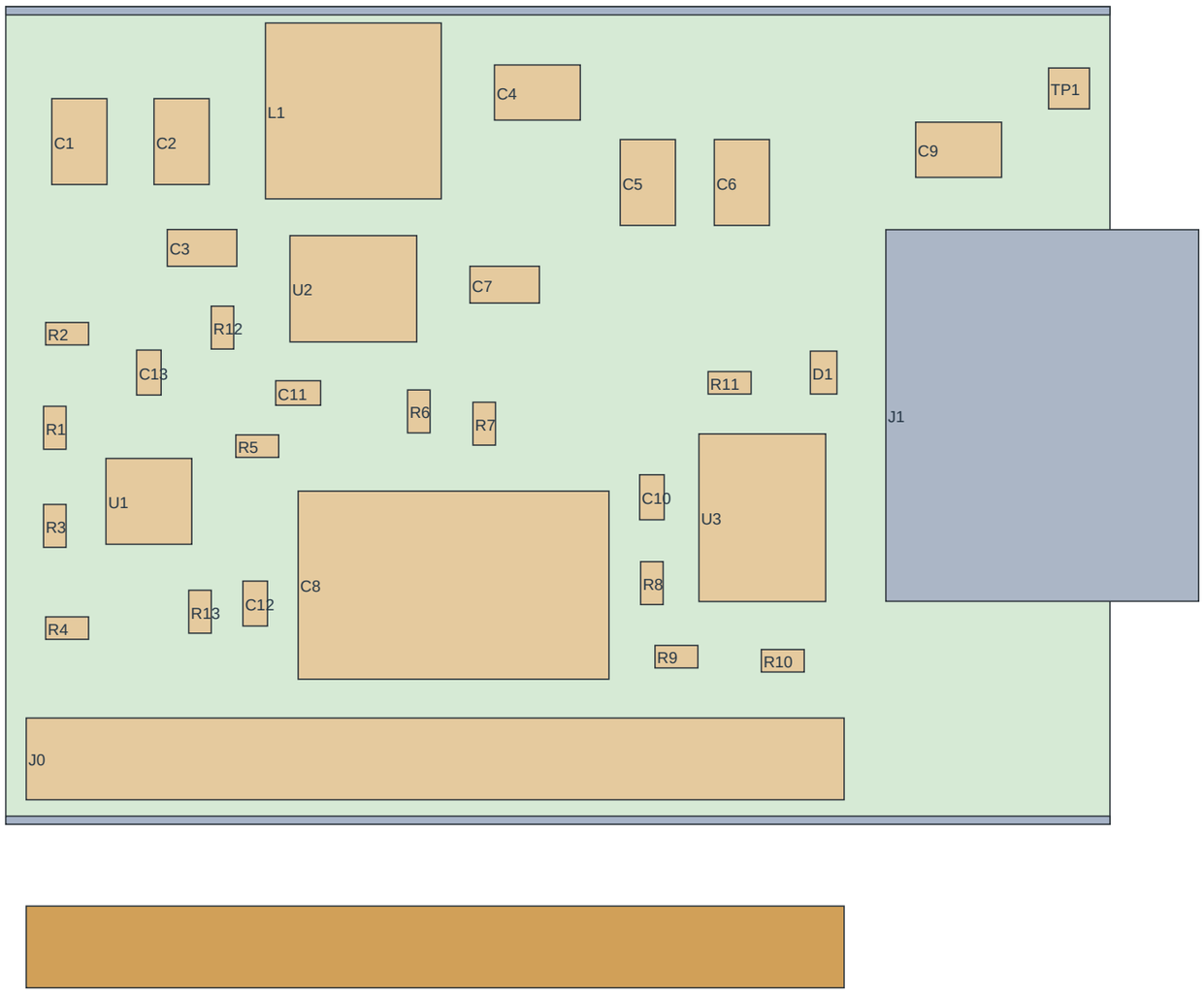
AXIAL ENVELOPE / X scale 10; radial scale 13 drawing units per mm



OD66. Locked X31..104.17 =73.17; released X26.8..104.17 =77.37. Core Ø48 x142 unchanged.

Connector front is a conservative body allocation, NOT a sourced mating plane. 12 x6.5 x12 plug screen is engineering-only.

ACTUAL SOURCE PLAN / local PCB27 x20; 18 drawing units per mm



External clamp / shaped flex tongue; screws outside routed board

Proper rotation: local +Z -> global +Y; local +Y -> global -Z.

Global=(75+x, 26.1+z, 10-y); determinant +1.

PCB X75..102 / Y25.1..26.1 / Z-10..10.

98 actual holes and four swept stake allocations retained.

Edge capture0.20; copper gap >=0.3005 from actual saved copper.

No new PCB holes. End retainer uses two external M1 screws.

Board1.0 +/-0.1; top lip clearance0.24 nominal.

Bottom reserve -1.35; top reserve +3.71 from PCB top.

R2.5 formed static bend; saddle ends before PCB edge.

Clamp shoe0.94 high; two M1 screws, preload unqualified.

Solder access slot20.8 x2.8; enclosure sealing/guard gate.

No insulating pad over ICs; no rated heat path asserted.

Four inherited GCT NPTH process errors remain unresolved.

Static flex, solder, spring, wear, thermal/current, source faults, USB profile and mains safety remain qualification gates.